CUSTOMER CHURN PREDICTION

Milestone 1 | Business Analysis and Proposed Modeling Approach

Prepared by: Timothy Mayabi | Email: usafiri2018@gmail.com

Data Science Capstone | Submitted: 21 October 2026

ACCOUNTS ANALYZED	CHURNED ACCOUNTS	OVERALL CHURN RATE
11,260	1,896	16.84%

Executive summary

Analysis of 11,260 customer accounts identifies 1,896 churned accounts (16.84%). Churn is concentrated in newly established accounts and in the Regular Plus segment. Accounts with tenure of 0–1 have an observed churn rate of 51.9%, while Regular Plus contributes 1,127 churned accounts, or 59.4% of all churn. These patterns suggest opportunities to investigate onboarding and segment-specific engagement, but they do not establish the causes of churn.

Business decision: Prioritize validation of early-tenure and Regular Plus risks, then test targeted interventions using incremental margin and campaign costs rather than offering blanket discounts.

1. Business problem and opportunity

In a competitive e-commerce market, losing an account may mean losing several customers associated with that account. The business requires a reliable way to identify at-risk accounts, understand observed churn patterns, and evaluate retention actions that preserve value without excessive incentives.

Project objective: Develop an account-level churn classifier and use predicted risk alongside account economics to inform financially defensible retention campaigns. Milestone 1 establishes the analytical evidence and methodology; no trained-model performance or campaign uplift is claimed here.

2. Dataset overview and limitations

The supplied Excel workbook contains a cross-sectional account-level dataset with 11,260 rows and 19 fields (including AccountID and binary Churn). Features include account tenure, city tier, contact frequency, payment method, satisfaction scores, account size and segment, complaints, revenue, cashback, and login device. The source does not document sampling dates, collection frequency, churn observation horizon, or the currency/units of revenue. These must be confirmed before deployment or monetary ROI estimates.

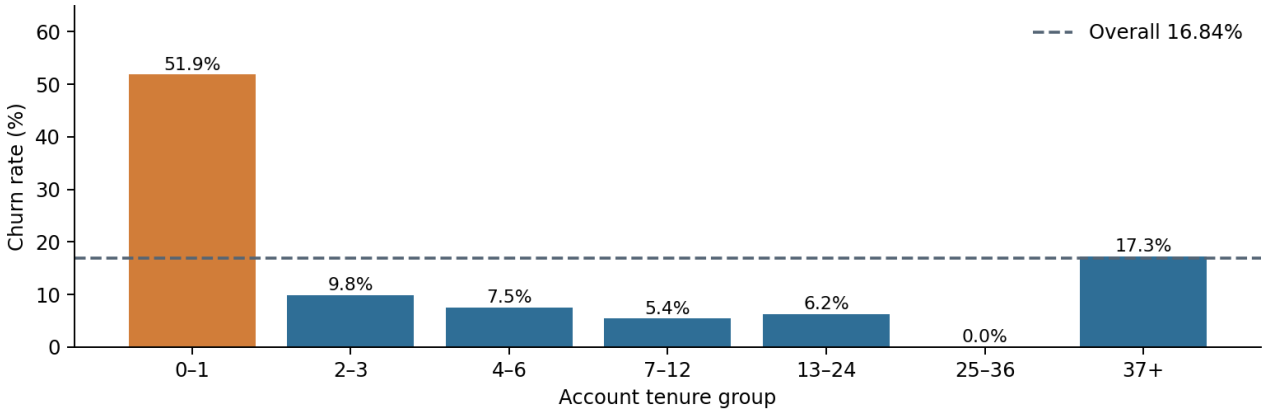
3. Data quality and preparation

The exploration found no duplicate records or duplicate AccountIDs. Seven fields expected to be numeric initially contained nonnumeric placeholders (for example #, $, +, *, and @); invalid entries were coerced to missing for analysis. Missingness after conversion was highest in rev_per_month (7.02%), cashback (4.20%), and Account_user_count (3.94%). Categorical values required normalization, including Regular + / Regular Plus, Super + / Super Plus, abbreviated gender values, and unrecognized device labels.

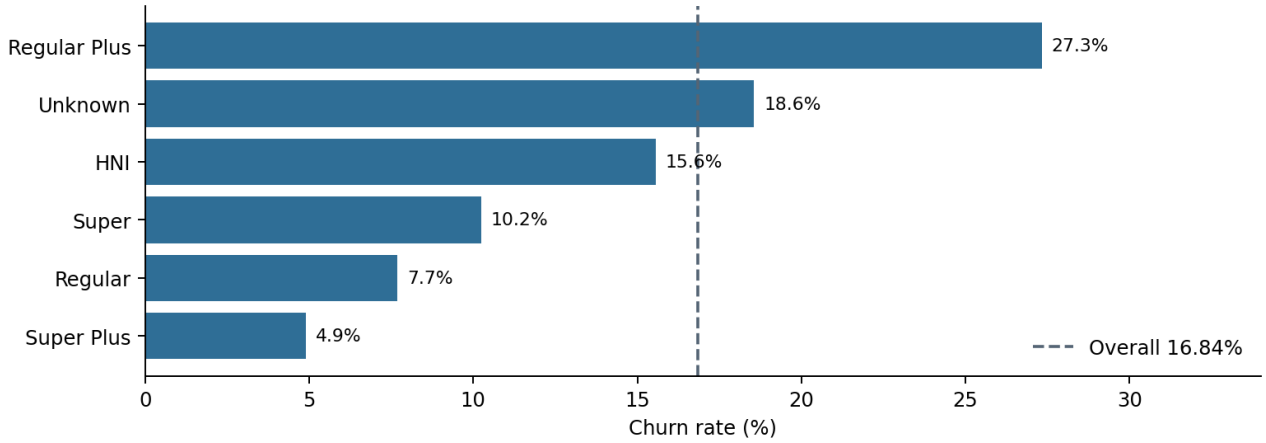
For modeling, missing-value imputation, categorical encoding, scaling where appropriate, and any outlier treatments will be fitted within training folds only. Extreme observations (including tenure 99 and unusually high revenue/cashback values) will be validated and tested for sensitivity rather than automatically deleted.

4. Exploratory data analysis — principal findings

Finding 1 | Early-tenure churn: The 0–1 tenure group has a 51.9% churn rate, approximately 3.1 times the overall rate. Median tenure is 1 among churned accounts versus 10 among retained accounts; Spearman correlation with churn is –0.401. Tenure units should be verified.



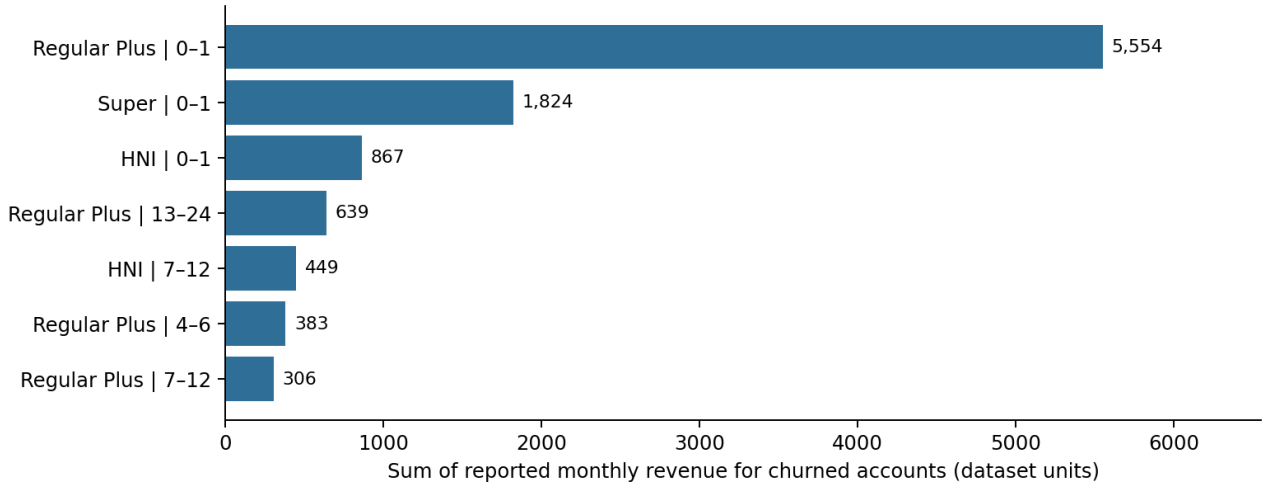
Finding 2 | Segment concentration: Regular Plus has a 27.33% churn rate and 1,127 churned accounts, representing 59.4% of total observed churn. High churn rate and large population make this segment an important investigation priority.



Finding 3 | Payment and care signals: Cash on Delivery (25.05%) and E-wallet (22.68%) have higher churn rates than Credit Card (14.21%) and Debit Card (15.30%). Median customer-care contacts are 17 for churned accounts versus 16 for retained accounts. These associations may be confounded by account tenure and segment.

5. Revenue exposure and commercial interpretation

The chart below ranks selected segment–tenure cohorts by the sum of reported monthly revenue associated with churned accounts. These are descriptive amounts in dataset units, not verified currency losses, profit, or recoverable revenue. They cannot establish the value of a campaign without margin, expected remaining lifetime, and causal intervention effects.



Illustrative priority: Regular Plus accounts in the 0–1 tenure group comprise 1,727 accounts, including 917 churned accounts (53.10%). The associated churned-account monthly revenue sum is 5,874 dataset units. This cohort combines substantial churn incidence and reported revenue exposure, but financial eligibility must be evaluated account by account.

6. Proposed model-building methodology

Stage	Proposed approach
Target and split	Predict Churn (1 = churned). Use stratified 80/20 train/test split; reserve test set for final assessment.
Preprocessing	Drop AccountID; impute missing values; normalize categories; one-hot encode nominal features; scale when required; use pipelines to avoid leakage.
Models	Logistic Regression baseline, Decision Tree, Random Forest, and XGBoost.
Validation	Stratified five-fold cross-validation; tune models on training folds; compare minority-class recall, precision, F1, PR-AUC, ROC-AUC, and calibration.
Decision threshold	Choose using validation predictions and estimated business costs, not an automatic 0.5 cutoff.
Interpretation	Use coefficients, permutation importance, and SHAP where appropriate; inspect stability by tenure and segment.

With 16.84% churn prevalence, accuracy alone would be misleading. Class weighting will be considered first; any oversampling will be performed only inside training folds. Potential target leakage and feature availability at prediction time will be reviewed before fitting models.

7. Proposed retention strategy and revenue assurance

Investigate: (a) early-tenure onboarding and proactive support; (b) Regular Plus account experience; (c) payment friction and service recovery. Avoid automatic blanket discounts. For each candidate intervention, estimate incremental retained contribution margin minus contact, incentive, and operational costs. Run randomized holdout

tests to estimate actual incremental retention effects.

Illustrative decision rule: Expected net benefit = probability of incremental retention due to intervention × expected retained contribution margin – total intervention cost. Churn probability is useful for targeting but is not the probability that an offer will work.

8. Action plan for Milestone 2

Phase	Deliverable
Data readiness	Validated definitions, data dictionary, leakage checks, preprocessing pipeline
Baseline and advanced modeling	Cross-validated Logistic Regression, Decision Tree, Random Forest, XGBoost
Evaluation and interpretation	Held-out test performance, calibrated risk scores, key predictive drivers
Campaign design	Segment-level targeting, unit economics, controlled test plan
Executive decision	Prioritized actions, estimated uncertainty, and go/no-go criteria

9. Risks, assumptions, and decisions needed

Data provenance: Confirm source period, sampling design, tenure units, churn definition and prediction horizon. **Financial validity:** Confirm rev_per_month and cashback units, margin, expected customer lifetime, and whether reported revenue is recognized revenue. **Data quality:** Investigate placeholder values, unknown categories, and extreme records. **Operational validity:** Confirm that every candidate feature is available before churn occurs. **Intervention evidence:** Obtain experimental or historical uplift estimates before recommending incentive amounts.

10. Conclusion

Milestone 1 identifies meaningful variation in observed account churn, especially for early-tenure and Regular Plus accounts. The next milestone will test predictive generalization and translate calibrated risk estimates into targeted interventions subject to revenue-assurance review. No causal effects, model performance claims, or guaranteed financial savings are asserted at this stage.

Source and analytical notes

Source: supplied “Customer Churn Data.xlsx”, worksheet “Data for DSBA” (11,260 rows). Descriptive statistics and charts are based on the provided workbook and the exploratory results reviewed in the project conversation. Rates are observed proportions; rounding may cause minor differences. Cohort comparisons are descriptive, not causal.